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# Engineering Dynamics: Particles and Rigid Bodies

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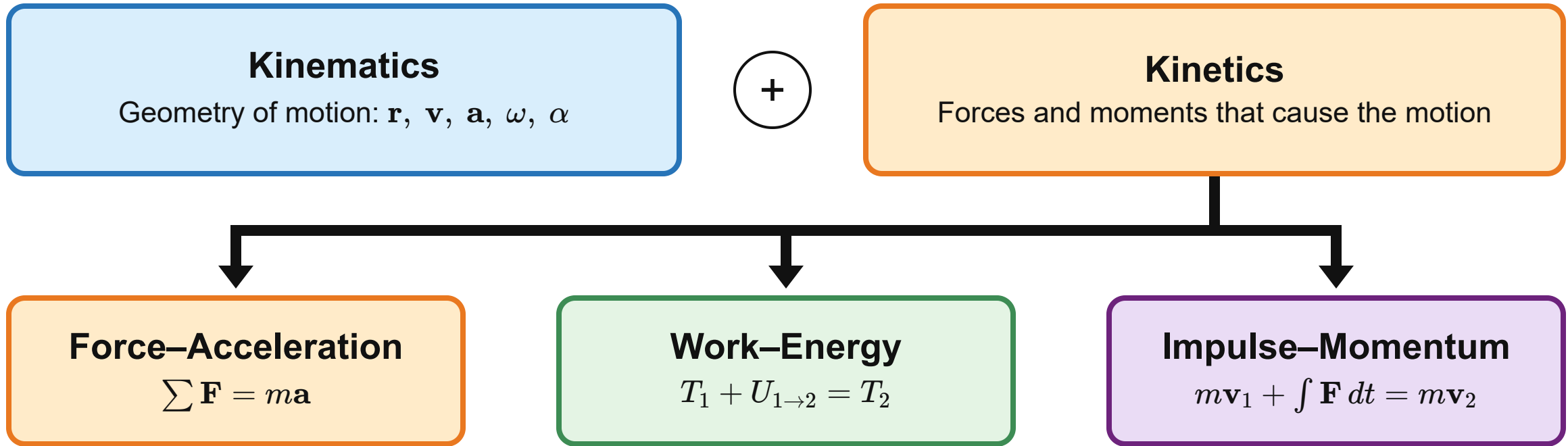
Core summary of J. L. Meriam & L. G. Kraige, *Engineering Mechanics: Dynamics*

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# Lecture Goals: What Dynamics Is About



**Dynamics = Kinematics (how a body moves) + Kinetics (why it moves)**



**Particle** : all mass concentrated at one point — size and rotation are ignored.

**Rigid body** : finite size, no deformation — **rotation must always be included**.

**This lecture** : apply kinematics and the three kinetics methods first to particles, then to rigid bodies.

# (Meriam) Dynamics: The Big Picture



The same four tools apply to both models — rigid bodies add rotation to each

| Topic              | Particle                          | Rigid Body                                                  |
|--------------------|-----------------------------------|-------------------------------------------------------------|
| Kinematics         | Position, velocity, acceleration  | Translation + rotation ( $\omega$ , $\alpha$ )              |
| Force–Acceleration | $\sum \mathbf{F} = m\mathbf{a}$   | $\sum \mathbf{F} = m\mathbf{a}_G$ , $\sum M_G = I_G \alpha$ |
| Work–Energy        | $T_1 + U_{1 \rightarrow 2} = T_2$ | Translational KE + rotational KE                            |
| Impulse–Momentum   | Linear momentum $m\mathbf{v}$     | Linear momentum + angular momentum $I_G \omega$             |

**The key difference** : a particle never rotates; a rigid body **always carries rotation** — new quantities  $\omega$ ,  $\alpha$ ,  $I$  and one extra equation,  $\sum M_G = I_G \alpha$ .

**Strategy** : master each tool on the particle first, then extend it by adding the rotational term.

# Particle Kinematics: Rectilinear Motion



Velocity and acceleration are successive time derivatives of position

## General definitions

$$\mathbf{v} = \frac{d\mathbf{r}}{dt}, \quad \mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}$$

## Straight-line motion

$$v = \frac{dx}{dt}, \quad a = \frac{dv}{dt}$$

$$a = v \frac{dv}{dx}$$

**Why**  $a = v dv/dx$ ? : when  $a$  is given as a function of position, this form eliminates time.

## Constant acceleration, $a = \text{const}$

$$v = v_0 + at$$

$$x = x_0 + v_0t + \frac{1}{2}at^2$$

$$v^2 = v_0^2 + 2a(x - x_0)$$

**Warning** : these three equations hold **only when  $a$  is constant**. For  $a(t)$ ,  $a(v)$  or  $a(x)$ , integrate the definitions directly.

# Curvilinear Motion: Cartesian and n-t Coordinates



Choosing the coordinate system is the key step in curvilinear problems

Cartesian (x-y)

$$\mathbf{v} = \dot{x} \mathbf{i} + \dot{y} \mathbf{j}, \quad \mathbf{a} = \ddot{x} \mathbf{i} + \ddot{y} \mathbf{j}$$

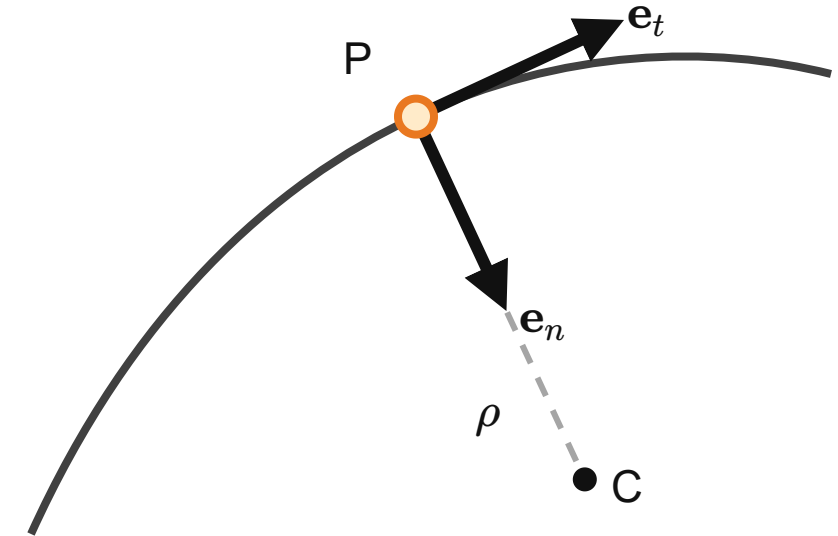
Normal-tangential (n-t)

$$\mathbf{a} = a_t \mathbf{e}_t + a_n \mathbf{e}_n, \quad a_t = \dot{v}$$

$$a_n = \frac{v^2}{\rho}$$

$$\text{Circular motion } (\rho = r) : a_n = \frac{v^2}{r} = \omega^2 r$$

**Meaning of  $a_n$  :** even at constant speed, a change of direction is an acceleration.



C : centre of curvature,  $\rho$  : radius of curvature

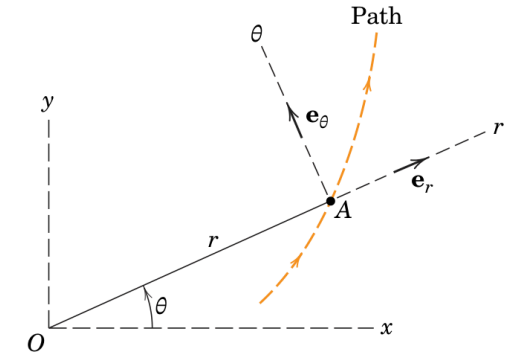
# Polar Coordinates: $r$ – $\theta$ Motion

For a body that moves while rotating, resolve along  $\mathbf{e}_r$  and  $\mathbf{e}_\theta$

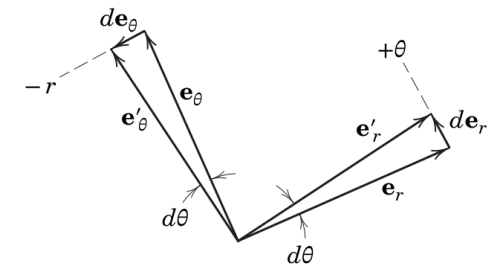
$$\mathbf{v} = \dot{r} \mathbf{e}_r + r\dot{\theta} \mathbf{e}_\theta$$

$$\mathbf{a} = \left( \ddot{r} - r\dot{\theta}^2 \right) \mathbf{e}_r + \left( r\ddot{\theta} + 2\dot{r}\dot{\theta} \right) \mathbf{e}_\theta$$

| Term                   | Direction       | Physical meaning                                      |
|------------------------|-----------------|-------------------------------------------------------|
| $\ddot{r}$             | radial          | Acceleration along the rotating line                  |
| $-r\dot{\theta}^2$     | radial (inward) | Centripetal acceleration from rotation                |
| $r\ddot{\theta}$       | transverse      | Angular acceleration of the line                      |
| $2\dot{r}\dot{\theta}$ | transverse      | Coriolis-type term — $r$ and $\theta$ change together |



(a)



(b)

FIGURE 2/13

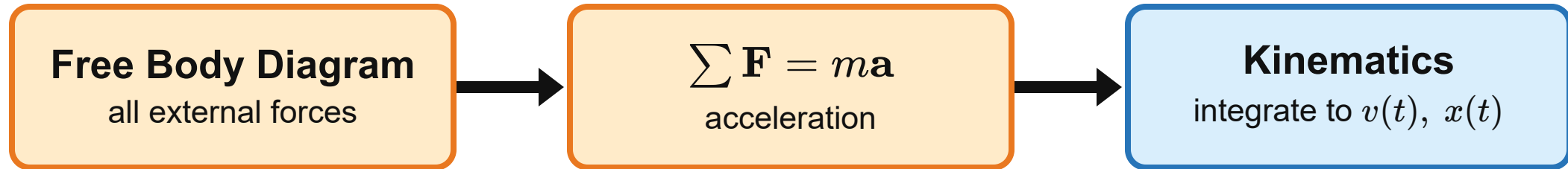
Meriam & Kraige, Fig. 2/13

**Example** : a bead sliding outward along a rotating rod — the rod must push it sideways, **even at constant  $\dot{\theta}$** , because of  $2\dot{r}\dot{\theta}$ .

# Particle Kinetics: Newton's Second Law



Force → acceleration → motion: every kinetics problem starts from a free body diagram



$$\sum \mathbf{F} = m\mathbf{a}$$

Cartesian :  $\sum F_x = m\ddot{x}$ ,  $\sum F_y = m\ddot{y}$

n-t components (curved paths)

$$\sum F_t = m\dot{v}, \quad \sum F_n = m\frac{v^2}{\rho}$$

**Habit** : draw the FBD before writing any equation — missing or duplicated forces are the most common source of error.

# Work–Energy Method



**Relates position and speed directly — no time variable needed**

$$T_1 + U_{1 \rightarrow 2} = T_2$$

**Kinetic energy and work**

$$T = \frac{1}{2}mv^2, \quad U_{1 \rightarrow 2} = \int_1^2 \mathbf{F} \cdot d\mathbf{r}$$

**Potential energy**

$$V_g = mgh, \quad V_e = \frac{1}{2}kx^2$$

**Only conservative forces**

$$T_1 + V_1 = T_2 + V_2$$

**Scalar equation** : no vector components and no reaction forces that do no work (normal force on a smooth path).

**When to use** : "find the speed, the distance is given, time is not needed" → **try energy before Newton's law.**

# Impulse–Momentum Method

Short, large forces and collisions are handled with momentum

$$m\mathbf{v}_1 + \int_{t_1}^{t_2} \sum \mathbf{F} dt = m\mathbf{v}_2$$

## Definitions

Linear momentum :  $\mathbf{p} = m\mathbf{v}$

Impulse :  $\int_{t_1}^{t_2} \mathbf{F} dt$

## No external impulse

$$m_1 v_1 + m_2 v_2 = m_1 v'_1 + m_2 v'_2$$

## Coefficient of restitution

$$e = \frac{v'_2 - v'_1}{v_1 - v_2} = \frac{\text{separation}}{\text{approach}}$$

$e = 1$  : perfectly elastic

$e = 0$  : perfectly plastic

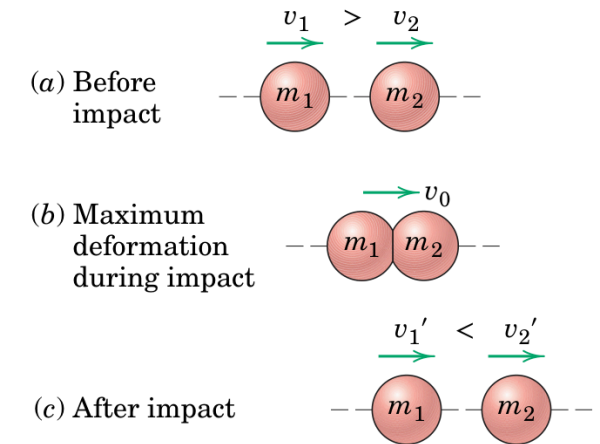


FIGURE 3/17

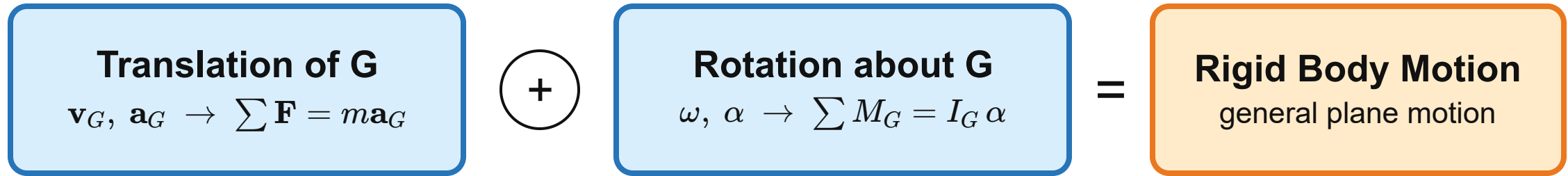
Meriam & Kraige, Fig. 3/17

**Collision recipe** : **momentum conservation + restitution** give two equations for  $v'_1$  and  $v'_2$ .

# Rigid Body Motion: Translation + Rotation



A rigid body moves its mass centre and rotates about it at the same time



## Translation ↔ rotation analogy

|             | Inertia | Velocity | Acceleration | Cause | Equation                | Kinetic energy            | Momentum     |
|-------------|---------|----------|--------------|-------|-------------------------|---------------------------|--------------|
| Translation | $m$     | $v$      | $a$          | $F$   | $\sum F = ma$           | $\frac{1}{2}mv^2$         | $mv$         |
| Rotation    | $I_G$   | $\omega$ | $\alpha$     | $M_G$ | $\sum M_G = I_G \alpha$ | $\frac{1}{2}I_G \omega^2$ | $I_G \omega$ |

**Reading the table** : every particle result has a rotational twin — replace  $m$  by  $I$ ,  $F$  by  $M$ ,  $a$  by  $\alpha$ .

# Rigid Body Kinematics: Relative Velocity



Velocity of B = velocity of A + rotation of B about A

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$$

**Two parts** : translation with the reference point A, plus circular motion of B about A.

$\mathbf{v}_{B/A} = \boldsymbol{\omega} \times \mathbf{r}_{B/A}$  : magnitude  $r\omega$ , perpendicular to  $\mathbf{r}_{B/A}$ .

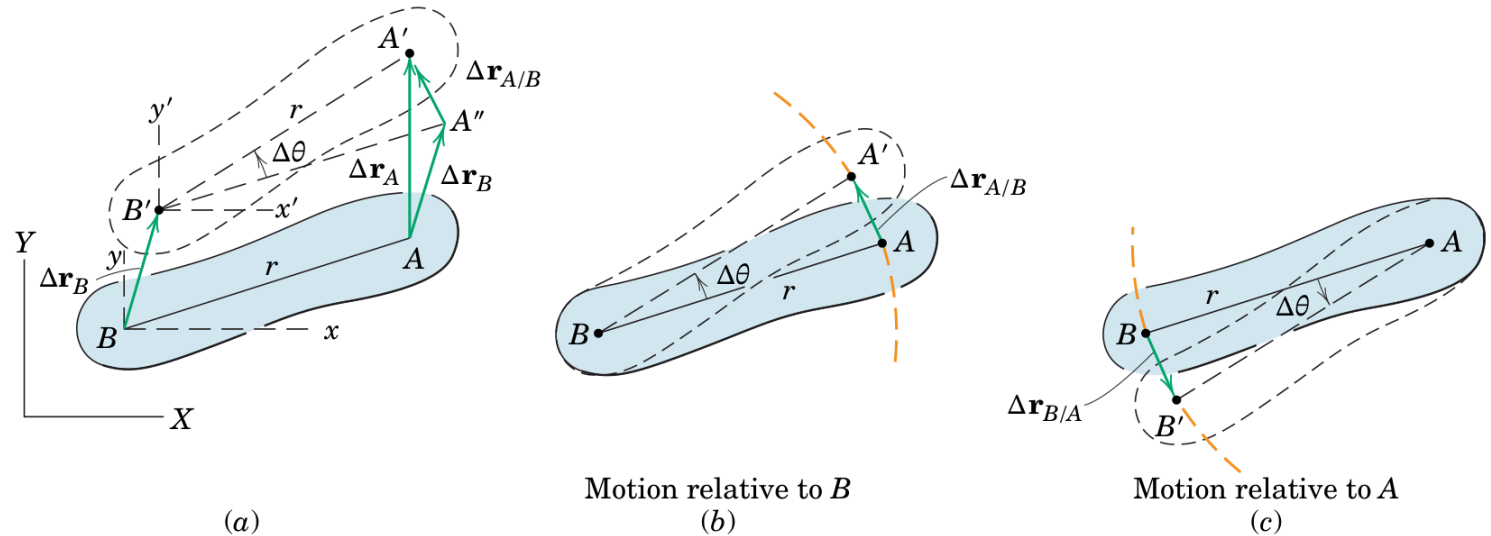


FIGURE 5/5

Meriam & Kraige, Fig. 5/5 — (a) displacement, (b) motion relative to B, (c) motion relative to A

**Why it matters** : this is **the most used equation in rigid-body kinematics** — linkages, wheels and gears all reduce to it.

# Rigid Body Kinematics: Relative Acceleration



Relative acceleration adds a tangential term  $r\alpha$  and a normal term  $r\omega^2$

$$\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{B/A} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{B/A})$$

## Plane motion

$$\mathbf{a}_B = \mathbf{a}_A + \boldsymbol{\alpha} \times \mathbf{r}_{B/A} - \omega^2 \mathbf{r}_{B/A}$$

| Term                                          | Magnitude   | Direction                                        |
|-----------------------------------------------|-------------|--------------------------------------------------|
| $\boldsymbol{\alpha} \times \mathbf{r}_{B/A}$ | $r\alpha$   | Tangential — perpendicular to $\mathbf{r}_{B/A}$ |
| $-\omega^2 \mathbf{r}_{B/A}$                  | $r\omega^2$ | Normal — from B toward A                         |

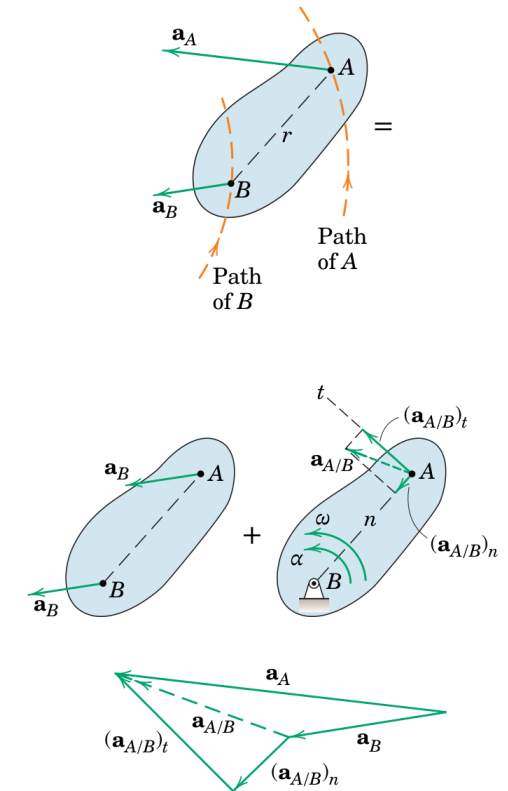


FIGURE 5/9

Meriam & Kraige, Fig. 5/9

**Solve it** : sketch the vector polygon — **known vectors first**, unknown magnitudes close the polygon.

# Three Types of Plane Motion



Classify the motion first — each type has its own simple relations

| Type of Rigid-Body Plane Motion |  | Example |
|---------------------------------|--|---------|
| (a)<br>Rectilinear translation  |  |         |
| (b)<br>Curvilinear translation  |  |         |
| (c)<br>Fixed-axis rotation      |  |         |
| (d)<br>General plane motion     |  |         |

FIGURE 5/1

Meriam & Kraige, Fig. 5/1

## Translation

(a), (b) —  $\omega = 0$

$$\mathbf{v}_A = \mathbf{v}_B, \quad \mathbf{a}_A = \mathbf{a}_B$$

treat the body as a particle

## Fixed-Axis Rotation

(c) — about fixed O

$$v = r\omega, \quad a_t = r\alpha, \quad a_n = r\omega^2$$

pulleys, flywheels, pendulums

## General Plane Motion

(d) — translation + rotation

translation of  $G$  + rotation about  $G$

wheels, links, connecting rods

**Priority** : general plane motion is the most important case — most exam problems (wheels, linkages, disks) belong here.

# Instantaneous Center of Zero Velocity



At each instant, a body in plane motion rotates about one point **C** where  $v_C = 0$

$$v_A = \omega r_{A/C}, \quad v_B = \omega r_{B/C}$$
$$\omega = \frac{v_A}{r_{A/C}} = \frac{v_B}{r_{B/C}}$$

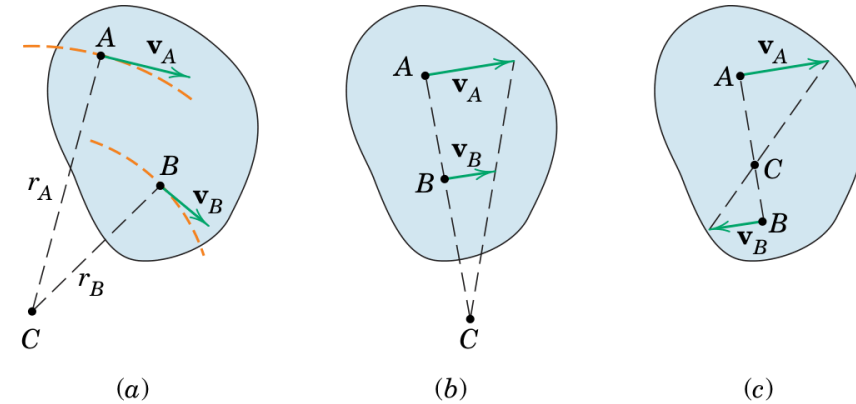


FIGURE 5/7

Meriam & Kraige, Fig. 5/7 — C lies on the normals to  $\mathbf{v}_A$  and  $\mathbf{v}_B$

**Locating C** : draw lines perpendicular to  $\mathbf{v}_A$  and  $\mathbf{v}_B$ ; they intersect at C (a). Parallel velocities: use their magnitudes (b), (c).

**Benefit** : the velocity of any point follows from one distance — no vector algebra.

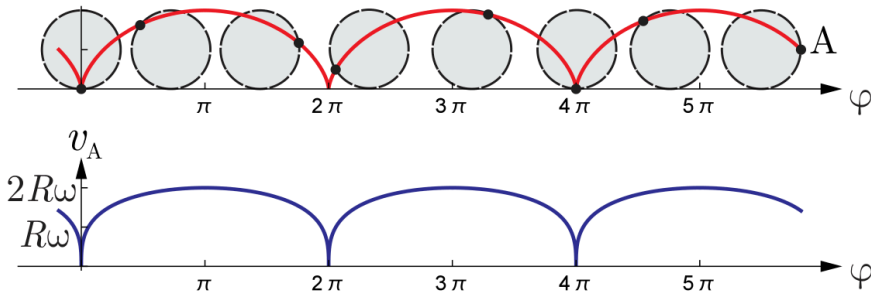
**Caution** : the IC is for **velocity only**. C itself generally accelerates, so  $a = \omega^2 r$  about C is wrong.

# Rolling Without Slipping



The contact point has zero velocity, which ties the motion of G to  $\omega$

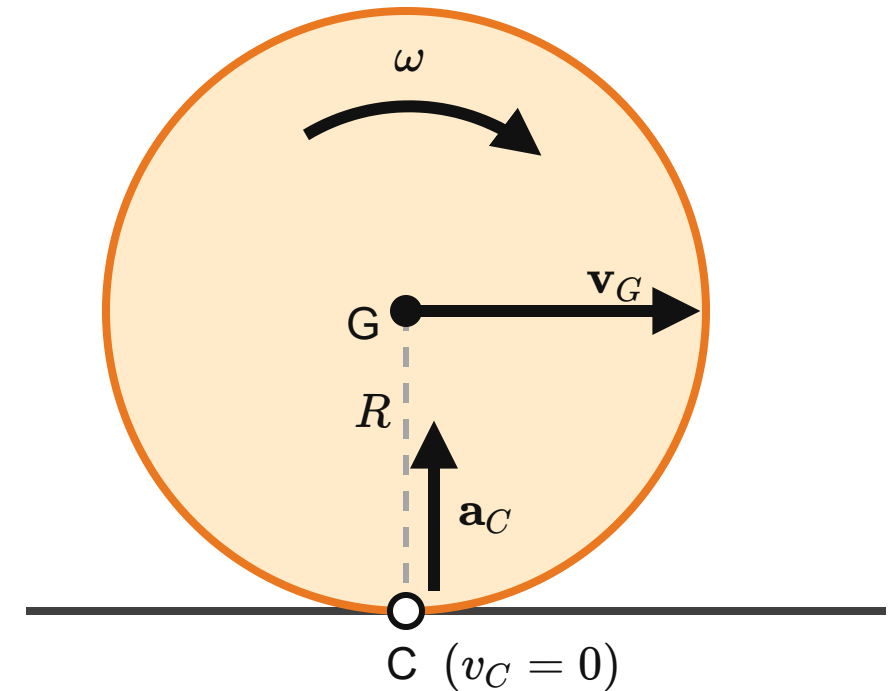
$$v_G = R\omega, \quad a_G = R\alpha$$



Kochmann, ETH Dynamics lecture notes, p. 84 — rim point traces a cycloid; its speed is 0 at contact,  $2R\omega$  at the top

**Common trap** : zero velocity at C does not mean zero acceleration. C accelerates toward G with  $a_C = R\omega^2$ .

**Friction** : static friction at C does no work, because C does not move at that instant.



# Mass Moment of Inertia



$I$  plays the role of mass in rotation — mass far from the axis counts more

## Definition

$$I = \int r^2 dm$$

## Parallel-axis theorem

$$I_O = I_G + md^2$$

$d$  : distance between the axis through G and the parallel axis through O.

## Common bodies

| Body        | Axis   | $I$                |
|-------------|--------|--------------------|
| Solid disk  | centre | $\frac{1}{2}mR^2$  |
| Thin ring   | centre | $mR^2$             |
| Slender rod | centre | $\frac{1}{12}mL^2$ |
| Slender rod | end    | $\frac{1}{3}mL^2$  |

**Quick check** : rod about its end,  $\frac{1}{12}mL^2 + m\left(\frac{L}{2}\right)^2 = \frac{1}{3}mL^2$  — the table's last row follows from the parallel-axis theorem.

# Rigid Body Kinetics: Force–Acceleration



Two equations used together — one for translation, one for rotation

|             |                                    |
|-------------|------------------------------------|
| Translation | $\sum \mathbf{F} = m \mathbf{a}_G$ |
| Rotation    | $\sum M_G = I_G \alpha$            |

Plane motion: three scalar equations

$$\sum F_x = m a_{Gx}, \quad \sum F_y = m a_{Gy}, \quad \sum M_G = I_G \alpha$$

Fixed-axis rotation about O

$$\sum M_O = I_O \alpha$$

**Procedure** : FBD  $\equiv$  kinetic diagram  $\rightarrow$  kinematic link (e.g.  $a_G = R\alpha$ )  $\rightarrow$  solve.

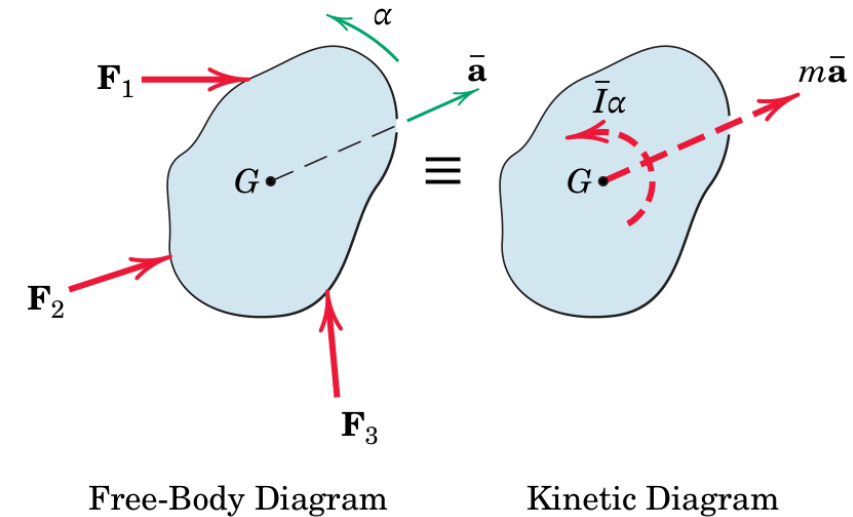


FIGURE 6/4

Meriam & Kraige, Fig. 6/4 — forces on the FBD  $\equiv m\bar{\mathbf{a}}$  and  $\bar{I}\alpha$  on the kinetic diagram

**Starting point** : almost every rigid-body kinetics problem starts from these two equations.

# Rigid Body Work–Energy



**Kinetic energy has two parts: translation of G and rotation about G**

$$T = \frac{1}{2}mv_G^2 + \frac{1}{2}I_G \omega^2, \quad T_1 + U_{1 \rightarrow 2} = T_2$$

**Example: solid disk rolling from rest down a height  $h$**

$$T = \frac{1}{2}mv^2 + \frac{1}{2} \left( \frac{1}{2}mR^2 \right) \left( \frac{v}{R} \right)^2 = \frac{3}{4}mv^2$$

$$mgh = \frac{3}{4}mv^2 \quad \Rightarrow \quad v = \sqrt{\frac{4}{3}gh} \quad \left( \text{sliding block: } v = \sqrt{2gh} \right)$$

**Rolling recipe** :  $T = \frac{1}{2}mv_G^2 + \frac{1}{2}I_G\omega^2$  together with  $v_G = R\omega$  solves most rolling problems.

**Interpretation** : rotation stores part of the energy, so the rolling disk arrives slower than a frictionless sliding block.

# Rigid Body Impulse–Momentum



Rigid bodies need both linear and angular momentum

|             |                                                                   |
|-------------|-------------------------------------------------------------------|
| Linear      | $m \mathbf{v}_{G1} + \int \sum \mathbf{F} dt = m \mathbf{v}_{G2}$ |
| Angular (G) | $I_G \omega_1 + \int \sum M_G dt = I_G \omega_2$                  |

No external moment impulse

$$I_1 \omega_1 = I_2 \omega_2$$

A spinning skater pulls in the arms:  $I$  drops, so  $\omega$  rises.

Typical uses

Impacts on bars and pendulums, spinning bodies that change shape, coupled rotors.

**Conservation** : with no external moment impulse, **angular momentum is conserved** — even when energy is not.

# Choosing the Method



**Pick the method before searching for a formula**

| What the problem gives or asks                    | Method                 | Start from                                                                  |
|---------------------------------------------------|------------------------|-----------------------------------------------------------------------------|
| Position, velocity, acceleration versus time      | Kinematics             | $v = dx/dt, a = dv/dt$                                                      |
| Forces linked directly to acceleration            | Force–Acceleration     | $\sum F = ma, \sum M_G = I_G \alpha$                                        |
| Position $\leftrightarrow$ speed, time not needed | Work–Energy            | $T_1 + U_{1 \rightarrow 2} = T_2$                                           |
| Impact, collision, large force over short time    | Impulse–Momentum       | $mv_1 + \int F dt = mv_2$                                                   |
| Velocities of several points on a rigid body      | Relative velocity / IC | $\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$ |
| Rolling                                           | Rolling constraint     | $v_G = R\omega$                                                             |
| Energy of a rigid body                            | Rigid-body KE          | $\frac{1}{2}mv_G^2 + \frac{1}{2}I_G\omega^2$                                |

**Core skill** : **choosing the method** matters more than recalling formulas — the right choice often removes an unknown entirely.

# Ten Equations to Remember



The minimum set to have ready before an exam

|   |                     |                                     |    |                         |                                                                             |
|---|---------------------|-------------------------------------|----|-------------------------|-----------------------------------------------------------------------------|
| 1 | Newton's 2nd law    | $\sum F = ma$                       | 6  | Relative acceleration   | $a_B = a_A + \alpha \times r_{B/A} + \omega \times (\omega \times r_{B/A})$ |
| 2 | Normal acceleration | $a_n = \frac{v^2}{\rho}$            | 7  | Rolling                 | $v_G = R\omega$                                                             |
| 3 | Work–energy         | $T_1 + U_{1 \rightarrow 2} = T_2$   | 8  | Rigid body, translation | $\sum F = ma_G$                                                             |
| 4 | Impulse–momentum    | $mv_1 + \int F dt = mv_2$           | 9  | Rigid body, rotation    | $\sum M_G = I_G \alpha$                                                     |
| 5 | Relative velocity   | $v_B = v_A + \omega \times r_{B/A}$ | 10 | Rigid-body KE           | $T = \frac{1}{2}mv_G^2 + \frac{1}{2}I_G\omega^2$                            |

**Pattern** : 1–4 are particle tools; 5–10 add rotation to the same ideas.



## Particle dynamics is $F = ma$ seen through motion, energy and momentum

- Kinematics describes motion; kinetics explains it through forces and moments  
→ Choose coordinates to fit the path: Cartesian, n–t, or r– $\theta$
- Three kinetics methods for a particle  
→  $\sum F = ma \cdot T_1 + U_{1 \rightarrow 2} = T_2 \cdot mv_1 + \int F dt = mv_2$
- Rigid-body dynamics adds **rotation**:  $\omega$ ,  $\alpha$ ,  $I$  and  $\sum M_G = I_G \alpha$   
→ Kinematics:  $\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$ , IC, rolling  $v_G = R\omega$   
→ Energy and momentum each gain a rotational term:  $\frac{1}{2}I_G\omega^2$ ,  $I_G \omega$
- Decide the method first, then write the equation